

```
In [3]: import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline

from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
```

```
In [7]: data = pd.read_csv('CH.SC.U4CSE24156_EX-11.4_1(PCA).csv')

print(data.shape)
print(data.head())
```

(150, 12)

	Property_ID	LotArea	OverallQual	YearBuilt	GrLivArea	FullBath	\
0	1	11770	5	1979	2516	1	
1	2	5360	7	1991	2458	2	
2	3	9890	9	1966	1269	1	
3	4	17918	8	2011	2046	2	
4	5	9691	5	2000	2595	1	

	BedroomAbvGr	GarageCars	GarageArea	DistanceFromCity	SalePrice	Sold
0	4	0	112	8.3	186739	1
1	5	3	139	11.0	388105	0
2	4	0	558	6.3	308009	1
3	5	2	0	27.3	326028	1
4	5	0	89	17.9	277980	1

```
In [9]: data.info()
```

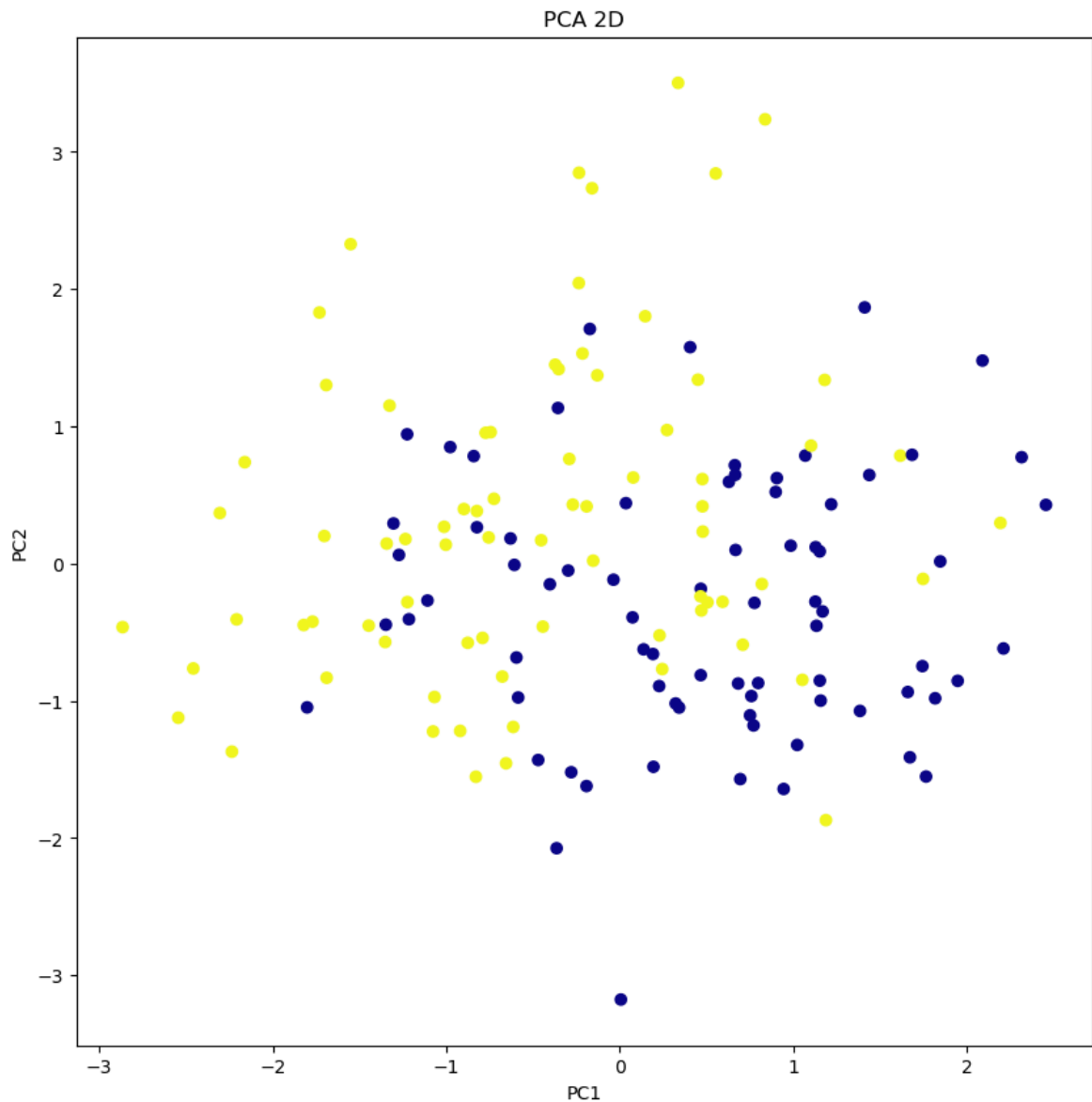
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Property_ID           150 non-null    int64
1   LotArea               150 non-null    int64
2   OverallQual           150 non-null    int64
3   YearBuilt             150 non-null    int64
4   GrLivArea             150 non-null    int64
5   FullBath              150 non-null    int64
6   BedroomAbvGr          150 non-null    int64
7   GarageCars            150 non-null    int64
8   GarageArea            150 non-null    int64
9   DistanceFromCity      150 non-null    float64
10  SalePrice              150 non-null    int64
11  Sold                  150 non-null    int64
dtypes: float64(1), int64(11)
memory usage: 14.2 KB
```

```
In [11]: X = data[
    [
        'LotArea',
        'OverallQual',
        'YearBuilt',
        'GrLivArea',
        'FullBath',
        'BedroomAbvGr',
        'GarageCars',
        'GarageArea',
```

```
        'DistanceFromCity',  
        'SalePrice'  
    ]  
]  
  
y = data['Sold']
```

```
In [13]: sc = StandardScaler()  
  
sc.fit(X)  
  
scaled_data = sc.transform(X)  
  
principal = PCA(n_components=3)  
  
principal.fit(scaled_data)  
  
x = principal.transform(scaled_data)  
  
print(x.shape)  
  
(150, 3)
```

```
In [15]: principal.components_  
  
plt.figure(figsize=(10,10))  
  
plt.title('PCA 2D')  
  
plt.scatter(  
    x[:,0],  
    x[:,1],  
    c=data['Sold'],  
    cmap='plasma'  
)  
  
plt.xlabel('PC1')  
plt.ylabel('PC2')  
  
plt.show()
```



```
In [17]: from mpl_toolkits.mplot3d import Axes3D

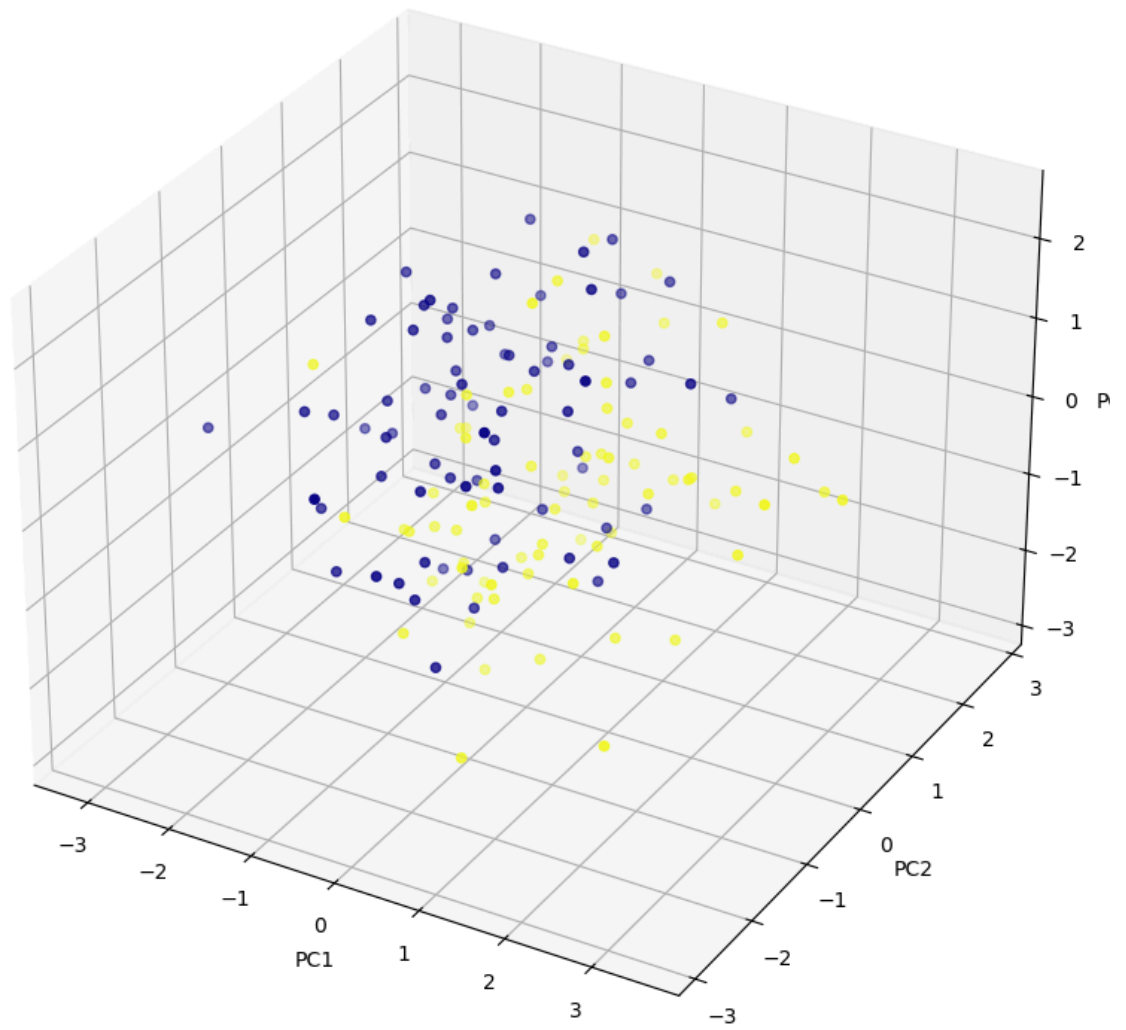
fig = plt.figure(figsize=(10,10))

ax = fig.add_subplot(111, projection='3d')

ax.scatter(
    x[:,1],
    x[:,2],
    x[:,0],
    c=data['Sold'],
    cmap='plasma'
)

ax.set_xlabel('PC1')
ax.set_ylabel('PC2')
ax.set_zlabel('PC3')

plt.show()
```



```
In [19]: print(principal.explained_variance_ratio_)
```

```
[0.13116519 0.12145932 0.117987  ]
```